01

OUR WORLD IN DATA VACCINATION EVIDENCE

The Power of Vaccination

From smallpox eradication to COVID-19 vaccine licensure in record time — a 1796–2020 view of public health progress.

Source: Our World in Data vaccination brief and figures.

1796 → 2020

2+ centuries

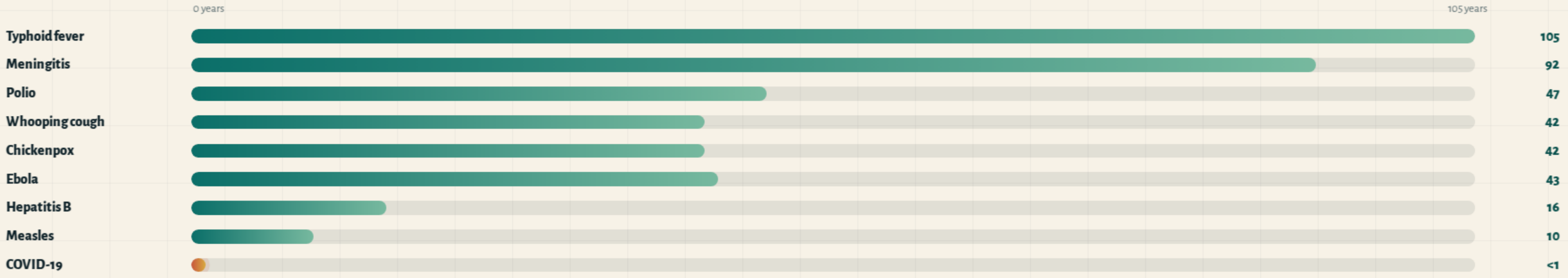
Vaccines moved from empirical discovery to genomic and mRNA-enabled development.

1796 1959 1980 2020

VACCINE-TIMELINE-ACCELERATION

From 100 years to under 1 year

Time from pathogen identification to vaccine licensure has shortened dramatically over 140 years.



Modern tools cited in the brief: advanced microscopy, genome sequencing, mRNA technology, and virus-like particles.

US-DISEASE-ELIMINATION

100% case reduction is achievable

US vaccine-preventable disease data show elimination for several diseases, with remaining burden concentrated in others.

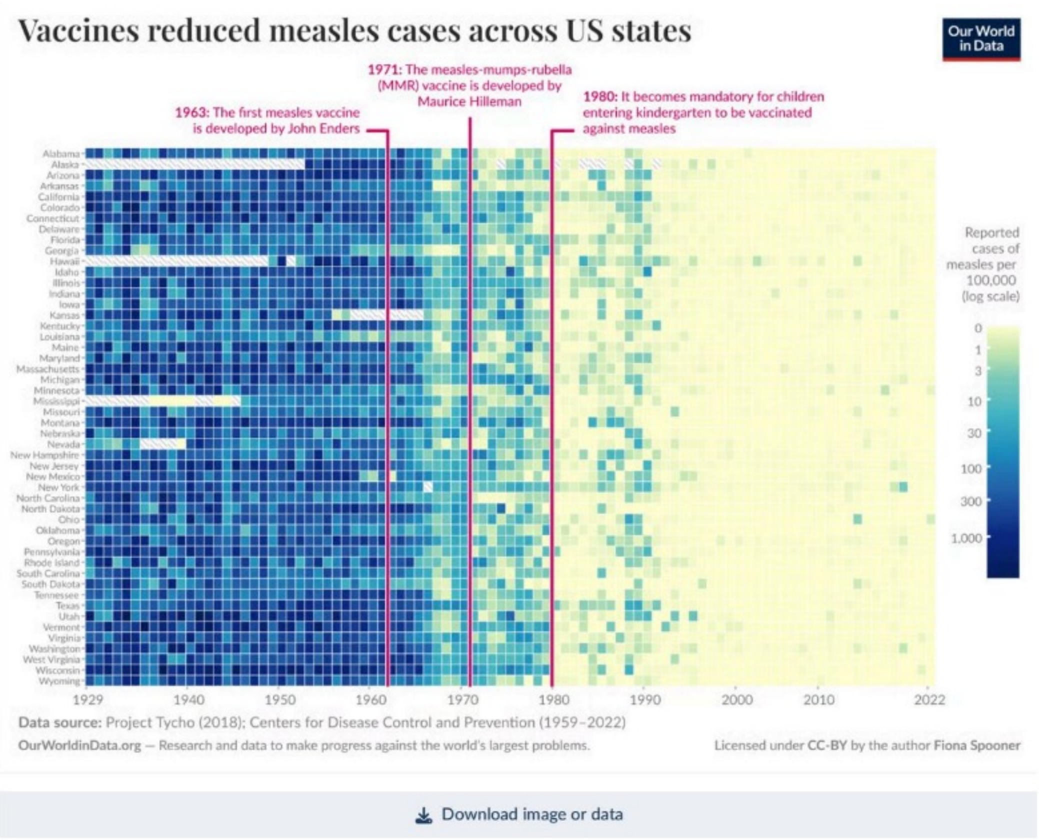
Disease	Pre-vaccine cases/M/yr	Post-vaccine cases/M/yr	Case reduction	Pre-vaccine deaths/M/yr	Death reduction
✓ Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
✓ Smallpox	250	0	100%	2.9	100%
✓ Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Data from Roush and Murphy (2007) as summarized in the source brief.

MEASLES-HEATMAP-STATES

Measles collapsed across all 50 states

The heatmap tracks measles cases per 100,000 in every US state from 1929 to 2022.



1963
First measles vaccine developed by John Enders.

1971
MMR vaccine developed by Maurice Hilleman.

1980
Vaccination became mandatory for children entering kindergarten.

Shift
From 100–1,000+ cases per 100,000 before 1963 to near zero after the 1980 mandate, with only sporadic outbreaks.

SMALLPOX-ERADICATION-STORY

The only human disease eradicated by vaccination

Smallpox was declared eradicated worldwide in 1980 after a coordinated global campaign.

1959
WHO resolution to eradicate smallpox.

1967
Intensified eradication campaign launched.

1977
Last natural case occurred in Somalia.

1980
Declared eradicated worldwide.

Why eradication was feasible

Human-only virus
No animal reservoir cited in the brief.

Distinguishable symptoms
Cases could be identified.

Cheap, effective vaccine
Practical for global campaign use.

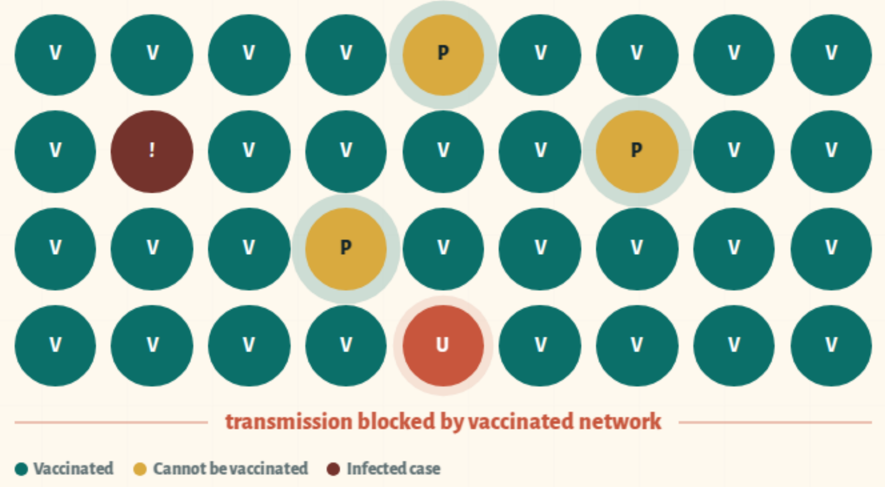
The brief identifies smallpox as the greatest vaccination success.

HERD-IMMUNITY-MECHANISM

Vaccinating many shields the most vulnerable

When enough people are vaccinated, pathogens cannot spread effectively — protecting people who cannot be vaccinated.

Protected groups: newborns, immunocompromised individuals, and cancer patients on chemotherapy.



REMAINING-CHALLENGES

Gaps persist where impact is incomplete

Not all vaccine-preventable disease burden has been eliminated; access, supply, and trust remain system barriers.

Pneumococcal

40.5%

Case reduction; death reduction is 31.3%.

Hepatitis A

89%

Case reduction; death reduction is 88.7%.

Acute hepatitis B

83.9%

Case reduction; death reduction is 83.6%.

System priorities from the brief: improve global access, reduce supply shortages, strengthen scientific understanding, and support uptake.

TAKEAWAYS-AND-PRIORITIES

Vaccines changed what public health can achieve

The evidence supports continued investment in access, coverage, trust, and newer vaccines.

1

Elimination is possible

Diphtheria, smallpox, and acute polio reached 100% US case and death reduction.

2

Timelines collapsed

Development moved from typhoid's 105 years to COVID-19 in under 1 year.

3

Mandates mattered

Measles dropped to near zero across all 50 states after the 1980 kindergarten mandate.

4

Communities benefit

Herd immunity shields newborns, immunocompromised people, and chemotherapy patients.

5

The job is unfinished

Pneumococcal disease, hepatitis A, and hepatitis B show persistent gaps.

Call to action: build global health systems that deliver vaccines reliably, equitably, and with public trust.

Access • Coverage • Trust